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Kerala, India
Portfolio

Mohammad Niyas

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Backend Engineer — Go (Golang)

GitHub
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Backend Engineer specializing in Go (Golang), PostgreSQL, and Redis, with strong DSA fundamentals and hands-on experience building high-throughput, concurrent distributed systems. Skilled in designing for consistency and resilience under heavy load — including Redis-backed distributed locking, atomic rollback compensation, and event-driven architectures using the Transactional Outbox pattern with RabbitMQ. Experienced in developing decoupled microservices via gRPC and Protocol Buffers, containerized with multi-stage Docker builds and automated CI/CD pipelines. Driven by writing clean, test-driven, maintainable backend code and engineering robust production platforms.

SKILLS

Language & Core Backend	Go (Golang), Goroutines & Channels, Sync Primitives, Context & Cancellation, Memory Profiling, SQL, Bash
API & Security	Gin, REST APIs, GraphQL, gRPC, Protocol Buffers, WebSockets, API Gateway & Routing, Rate Limiting, Service-to-Service Communication
Databases & Cache	RESTful API Contracts, RBAC, JWT, Access & Refresh Tokens, OAuth 2.0, Webhooks, HMAC-SHA256 Signature Verification
Message Queues	PostgreSQL, Redis, MongoDB
Cloud & DevOps	RabbitMQ, Transactional Outbox Pattern, Async Event Publishing
Architecture	AWS (EC2, S3, IAM, EKS), Docker, Kubernetes, NGINX, GitHub Actions CI/CD, Linux
Testing & QA	Clean Architecture, SOLID Principles, Microservices, Event-Driven Architecture, Double-Entry Ledger Design
	Unit & Integration Testing, Table-Driven Testing, Postman, Swagger

PROJECTS

BookMyVenue — Distributed Venue Reservation & Transaction Engine



Nov 2025 — Present

- Engineered a 10-minute distributed inventory-hold engine using Redis SET NX with TTL and rollback compensation, eliminating double-booking collisions under high concurrent checkout traffic without ever locking the primary database.
- Designed a 2-stage (30% advance / 70% final) payment workflow integrated with Razorpay, guaranteeing financial integrity via HMAC-SHA256 signature verification and atomic 4-table PostgreSQL transactions with append-only audit logging for full traceability.
- Implemented an in-memory Token Bucket rate limiter in Redis using custom Lua scripts with fail-open safety, protecting public authentication and booking APIs against brute-force attacks and bot scrapers with sub-millisecond overhead.
- Implemented a GraphQL admin dashboard using gqlgen with typed resolvers for venue approval workflows, user management (suspend/ban), booking oversight, and real-time platform analytics with aggregate queries.
- Architected a shadow-copy draft system (VenueEditDraft) allowing venue owners to submit major edits for admin review without corrupting or unpublishing the live production listing.
- Built a transactional outbox worker polling PostgreSQL with FOR UPDATE SKIP LOCKED, publishing booking events to RabbitMQ with retry logic and dead-letter handling, decoupling notification delivery from the booking critical path.

VogueLuxe — Cloud-Native Microservices E-Commerce Platform



Feb 2025 — Jun 2025

- Refactored a monolithic backend by extracting high-velocity domains (Wallet & Wishlist) into standalone gRPC microservices using Protocol Buffers, achieving strict fault isolation and sub-millisecond inter-service RPC latency.
- Built a reverse-proxy API Gateway handling centralized JWT authentication, role-based access control (RBAC), and protocol translation from incoming client HTTP requests to internal gRPC invocations.
- Designed a double-entry wallet ledger enforcing strict credit-debit balance consistency across asynchronous Razorpay payment webhooks, preventing fund mismatches during retries or out-of-order events.
- Containerized all services with multi-stage Docker builds to minimize image size, and automated CI/CD deployment via GitHub Actions into an AWS EKS Kubernetes cluster with zero-downtime rolling updates.

EDUCATION & TRAINING

Bachelor of Computer Applications (BCA) | Manipal University (Online)

2026 — Expected 2029

- Pursued via flexible online mode; 100% available for immediate, full-time employment.

Backend Engineering Residency | Brototype, Kerala

Sep 2024 — Present

- Intensive 20-month, project-based engineering program covering Go backend development, advanced DSA, microservices architecture, cloud deployments (AWS, Docker, Kubernetes), and weekly technical code reviews.